Operating system and system programming (25CS2104E)

TITLE

Linux IPC:
Pipes & FIFOs

Review-I

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Contents

- Abstract
- Introduction
- Hardware and Software requirements
- System Architecture
- Conclusion

Abstract

Inter-Process Communication (IPC) is a mechanism that allows processes to exchange data and coordinate their activities. In Linux, **Pipes and FIFOs** are simple and efficient IPC mechanisms used for communication between processes.

A **Pipe** provides a unidirectional communication channel, generally between related processes such as a parent and child process. A **FIFO (First-In, First-Out)**, also called a named pipe, provides communication between processes that do not necessarily have a parent-child relationship.

This project demonstrates how Linux Pipes and FIFOs can be used to transfer data between processes using system calls and file operations.

Introduction

Inter-Process Communication (IPC)

•IPC enables two or more processes to **exchange information and synchronize their activities**.

•Linux provides several IPC mechanisms such as:

- Pipes
- FIFOs
- Message Queues
- Shared Memory
- Sockets
- Signals

Pipes

•A pipe is an IPC mechanism that creates a **communication channel between processes**.

•Data written into one end of the pipe can be read from the other end.

•Pipes are generally **unidirectional**.

•They are commonly used between related processes.

FIFOs

•FIFO stands for **First-In, First-Out**.

•It is also known as a **Named Pipe**.

•Unlike an ordinary pipe, a FIFO has a **name in the file system**.

•It can be used for communication between unrelated processes.

Objective

To understand and demonstrate the working of **Linux Pipes and FIFOs** for inter-process communication.

Hardware and Software requirements

Hardware and Software Requirements

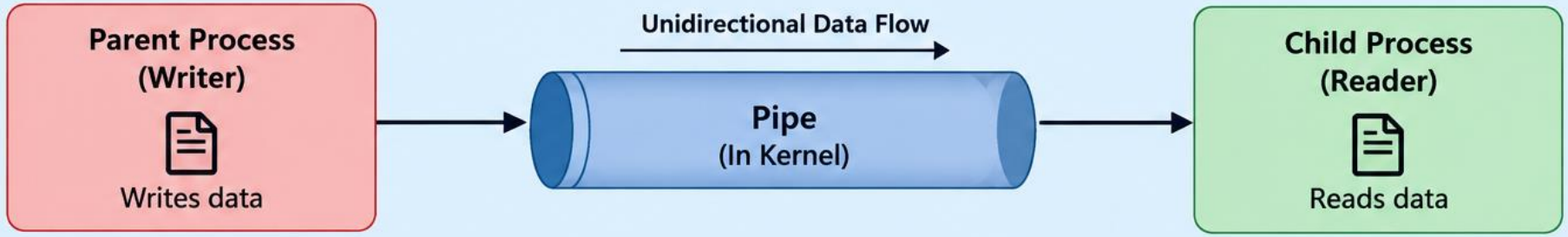
Requirement Type	Requirement
Processor	Intel/AMD Processor
RAM	Minimum 4 GB
Storage	Minimum 10 GB Free Space
Operating System	Linux / Ubuntu
Programming Language	C
Compiler	GCC
IPC Mechanisms	Pipes and FIFOs
System Calls	pipe(), read(), write(), close()
FIFO Functions	mkfifo(), open(), read(), write()
Development Environment	GCC / Linux Terminal
Libraries	Standard C and POSIX libraries

System Architecture

Linux Inter-Process Communication (Pipes and FIFOs)

1. Pipe (Anonymous Pipe)

Unidirectional communication between related processes (e.g., parent and child).



2. FIFO (Named Pipe)

Communication between unrelated processes (no parent-child relationship required).



Key System Calls / Operations

pipe()	fork()	read()	write()	mkfifo()	open()
(Create pipe)	(Create related process)	(Read data)	(Write data)	(Create named pipe)	(Open FIFO)

Thank you